

$$|\phi_{out}\rangle = \hat{S} |\phi_{in}\rangle$$

$$|\phi_{in}\rangle = \int_{-\infty}^{+\infty} dp \, C_{in}(p) |\psi_p^F\rangle$$

$$|\psi\rangle = \int_{-\infty}^{+\infty} dp \, C_{in}(p) |\psi_p^+\rangle$$

$$= \int_{-\infty}^{+\infty} dp \, C_{out}(p) |\psi_p^-\rangle$$

$$|\phi_{out}\rangle = \int_{-\infty}^{+\infty} dp \, C_{out}(p) |\psi_p^-\rangle$$

$$|\psi_p^F\rangle = |\psi_p^{FREE}\rangle$$

LSE

$$|\psi_p^\pm\rangle = |\psi_p^F\rangle + \frac{1}{E_p - \hat{H}_0 \pm i\epsilon} \hat{V} |\psi_p^F\rangle \quad \text{EXPLICIT}$$

$$|\psi_p^\pm\rangle = |\psi_p^F\rangle + \frac{1}{E_p - \hat{H}_0 \pm i\epsilon} \hat{V} |\psi_p^\pm\rangle \quad \text{IMPLICIT}$$

$$|\phi_{out}\rangle = \underbrace{\hat{S}} |\phi_{in}\rangle$$

$$\langle \psi_p^F | \psi_{p'}^F \rangle = \delta(p-p') = \langle \psi_p^\pm | \psi_{p'}^\pm \rangle$$

$$\langle \psi_p^+ | \psi_{p'}^+ \rangle = \langle \psi_p^F | \underbrace{|\psi_{p'}^+\rangle}_{\text{IMPLICIT LSE}} \rangle + \langle \psi_p^F | \hat{V} \frac{1}{E_p - \hat{H}_0 \pm i\epsilon} |\psi_{p'}^+\rangle =$$

↑
EXPLICIT
LSE

$$= \underbrace{\langle \psi_p^F | \psi_{p'}^F \rangle}_{\delta(p-p')} + \langle \psi_p^F | \frac{1}{E_p - \hat{H}_0 \pm i\epsilon} \hat{V} |\psi_{p'}^+\rangle + \langle \psi_p^F | \hat{V} \frac{1}{E_p - \hat{H}_0 \pm i\epsilon} |\psi_{p'}^+\rangle =$$

$$\rightarrow \delta(p-p') + \langle \psi_p^F | \hat{V} |\psi_{p'}^+\rangle \left(\frac{1}{E_{p'} - E_p \pm i\epsilon} - \frac{1}{E_p \pm E_{p'} \pm i\epsilon} \right)$$

$$\langle \psi_p^- | \psi_{p'}^+ \rangle$$

$$E_p = \frac{p^2}{2m}$$

$$E_{p'} = \frac{p'^2}{2m}$$

$$|\phi_{out}\rangle = \hat{S} |\phi_{in}\rangle$$

$$|\phi_{in}\rangle = \int_{-\infty}^{+\infty} dp' C_{in}(p') |\psi_p^F\rangle$$

$$|\psi\rangle = \int_{-\infty}^{+\infty} dp' C_{in}(p') |\psi_p^+\rangle = \int_{-\infty}^{+\infty} dp C_{out}(p) |\psi_p^-\rangle$$

$$|\phi_{out}\rangle = \int_{-\infty}^{+\infty} dp C_{out}(p) |\psi_p^F\rangle$$

$$C_{out}(p) = \langle \psi_p^- | \psi \rangle$$

$$C_{in}(p') = \langle \psi_{p'}^F | \phi_{in} \rangle$$

$$|\psi_p^F\rangle = |\psi_p^{FREE}\rangle$$

LSE

$$|\psi_p^\pm\rangle = |\psi_p^F\rangle + \frac{1}{E_p - \hat{H} \pm i\varepsilon} \hat{V} |\psi_p^F\rangle \quad \text{EXPLICIT}$$

$$|\psi_p^\pm\rangle = |\psi_p^F\rangle + \frac{1}{E_p - \hat{H}_0 \pm i\varepsilon} \hat{V} |\psi_p^\pm\rangle \quad \text{IMPLICIT}$$

$$|\phi_{out}\rangle = \int_{-\infty}^{+\infty} dp C_{out}(p) |\psi_p^F\rangle = \int_{-\infty}^{+\infty} dp |\psi_p^F\rangle \langle \psi_p^- | \psi \rangle =$$

$$= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle \langle \psi_p^- | \psi_{p'}^+ \rangle C_{in}(p') =$$

$$= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle \langle \psi_p^- | \psi_{p'}^+ \rangle \langle \psi_{p'}^F | \phi_{in} \rangle$$

$$\hat{S} = \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle \langle \psi_p^- | \psi_{p'}^+ \rangle \langle \psi_{p'}^F |$$

$$\langle \psi_p^F | \hat{S} | \psi_{p'}^F \rangle = \langle \psi_p^- | \psi_{p'}^+ \rangle$$

S-MATRIX

$$\langle \psi_p^F | \psi_{p'}^F \rangle = \delta(p-p')$$

$$\langle \psi_p^- | \psi_{p'}^+ \rangle = \langle \psi_p^F | \psi_{p'}^+ \rangle + \langle \psi_p^F | \hat{V} \frac{1}{E_p - \hat{H} + i\varepsilon} |\psi_{p'}^+ \rangle =$$

↑ EXPLICIT LSE

↑ IMPLICIT LSE

$$= \underbrace{\langle \psi_p^F | \psi_{p'}^F \rangle}_{\delta(p-p')} + \langle \psi_p^F | \frac{1}{E_p - \hat{H}_0 + i\varepsilon} \hat{V} |\psi_{p'}^+ \rangle + \langle \psi_p^F | \hat{V} \frac{1}{E_p - E_{p'} + i\varepsilon} |\psi_{p'}^+ \rangle =$$

$$= \delta(p-p') + \langle \psi_p^F | \hat{V} |\psi_{p'}^+ \rangle \left(\frac{1}{E_p - E_{p'} + i\varepsilon} + \frac{1}{E_p - E_{p'} + i\varepsilon} \right)$$

$$\frac{1}{(E_p - E_{p'}) + i\varepsilon} - \frac{1}{(E_p - E_{p'}) - i\varepsilon} = \frac{(E_p - E_{p'}) - i\varepsilon - (E_p - E_{p'}) - i\varepsilon}{(E_p - E_{p'})^2 + \varepsilon^2} = \frac{-2i\varepsilon}{(E_p - E_{p'})^2 + \varepsilon^2} = -2\pi i \frac{1}{\pi} \frac{\varepsilon}{(E_p - E_{p'})^2 + \varepsilon^2}$$

$$E_p = \frac{p^2}{2m}$$

$$E_{p'} = \frac{p'^2}{2m}$$

$$\delta(E_p - E_{p'})$$

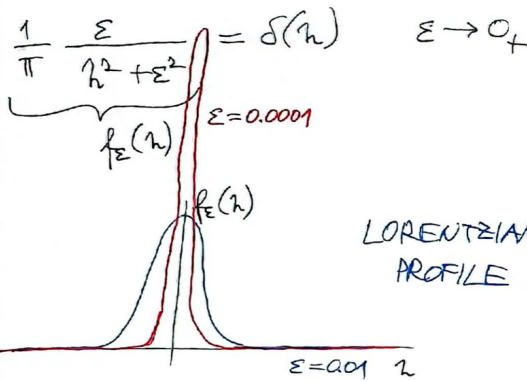
$$\langle \psi_p^F | \psi_{p'}^F \rangle = \delta(p-p')$$

$$\langle \psi_p^- | \psi_{p'}^+ \rangle = \delta(p-p') - 2\pi i \delta(E_p - E_{p'}) \langle \psi_p^F | \hat{V} | \psi_{p'}^F \rangle$$

$$E_p = \frac{p^2}{2m}$$

$$E_{p'} = \frac{p'^2}{2m}$$

$$\begin{aligned} \langle \psi_p^F | \hat{V} | \psi_{p'}^+ \rangle &= \langle \psi_p^F | \hat{V} | \psi_{p'}^F \rangle + \langle \psi_p^F | \hat{V} \frac{1}{E_p - \hat{H} + i\epsilon} \hat{V} | \psi_{p'}^F \rangle = \\ &= \langle \psi_p^F | \hat{V} + \hat{V} \frac{1}{E_p - \hat{H} + i\epsilon} \hat{V} | \psi_{p'}^F \rangle \end{aligned}$$



$$|\psi_p^F\rangle = |\psi_p^{\text{FREE}}\rangle$$

LSE

$$|\psi_p^\pm\rangle = |\psi_p^F\rangle + \frac{1}{E_p - \hat{H} \pm i\epsilon} \hat{V} |\psi_p^F\rangle \quad \text{EXPLICIT}$$

$$|\psi_p^\pm\rangle = |\psi_p^F\rangle + \frac{1}{E_p - \hat{H}_0 \pm i\epsilon} \hat{V} |\psi_p^\pm\rangle \quad \text{IMPLICIT}$$

SCATTERING MATRIX

$$E_p = \frac{p^2}{2m}$$

$$\langle \psi_p^F | \hat{S} | \psi_{p'}^F \rangle = \langle \psi_p^- | \psi_{p'}^+ \rangle =$$

ELASTIC ENERGY CONSERVATION IN SCATTERING

$$= \delta(p-p') - 2\pi i \delta(E_p - E_{p'}) \langle \psi_p^F | \hat{V} + \hat{V} \frac{1}{E_p - \hat{H} + i\epsilon} \hat{V} | \psi_{p'}^F \rangle$$

$$\langle \psi_p^F | \hat{S} | \psi_{p'}^F \rangle = 0$$

WHENEVER $E_p \neq E_{p'}$

TRANSITION MATRIX
ON-SHELL T-MATRIX

$\hat{T}$ -OPERATOR

$$\hat{T}(E) = \hat{V} + \hat{V} \frac{1}{E - \hat{H} + i\epsilon} \hat{V}$$

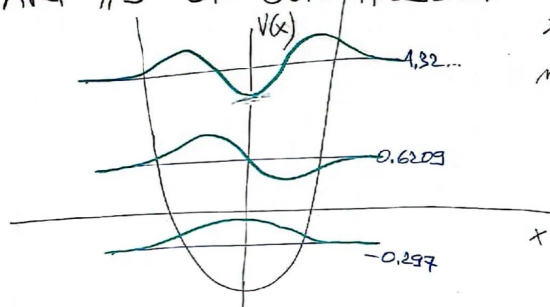
CONTAINS ALL

INFORMATION ABOUT
SCATTERING

BORN EXPANSION FOR THE $\hat{T}$ -OPERATOR

$$\hat{T}(E) = \hat{V} + \hat{V} \frac{1}{E - \hat{H}_0 + i\epsilon} \hat{V} + \hat{V} \frac{1}{E - \hat{H}_0 + i\epsilon} \hat{V} \frac{1}{E - \hat{H}_0 + i\epsilon} \hat{V} + \dots$$

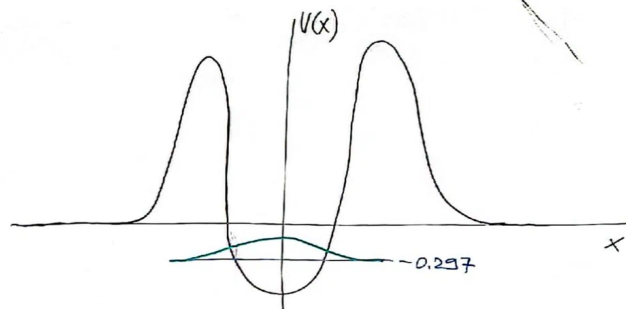
PART #5 OF OUR PROJECT



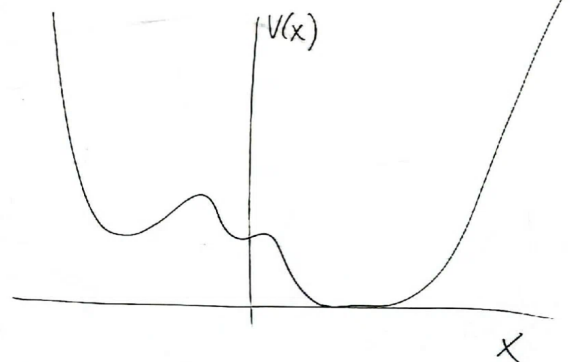
$$V(x) = 0.5x^2 - 0.8$$

$$\hat{H} = -\frac{\hbar^2}{2m} \partial_{xx} + V(x)$$

$$\hat{H} \psi_n(x) = E_n \psi_n(x)$$



$$V(x) = (0.5x^2 - 0.8)e^{-0.1x^2}$$



COMPLETE ORTHONORMAL BASIS SET ON $L^2(\mathbb{R})$

$$\{\phi_j(x)\}_{j=1}^{\infty}$$

$$\psi_m(x) = \sum_{j=1}^{\infty} c_{mj} \phi_j(x)$$

$$\langle \psi_m | \psi_m \rangle = \sum_{j=1}^{\infty} |c_{mj}|^2$$

$$\underline{\underline{H}} \quad \text{MATRIX WITH ELEMENTS} \quad \langle \phi_j | \hat{H} | \phi_{j'} \rangle$$

$$\vec{c}^{(n)} = \begin{pmatrix} c_{1n} \\ c_{2n} \\ \vdots \\ c_{jn} \end{pmatrix}$$

$$\hat{H} \sum_{j=1}^{\infty} c_{jm} \phi_j(x) = E_m \sum_{j=1}^{\infty} c_{jm} \phi_j(x) \quad / \quad \phi_j^*(x)$$

$$\sum_{j'=1}^{\infty} c_{j'm} \langle \phi_j | \hat{H} | \phi_{j'} \rangle = E_m \sum_{j'=1}^{\infty} c_{j'm} \underbrace{\langle \phi_j | \phi_{j'} \rangle}_{\delta_{jj'}} \int_{\mathbb{R}} dx$$

$$= E_m c_{jm}$$

$$\underline{\underline{H}} \vec{c}^{(n)} = E_n \vec{c}^{(n)}$$

IN PRACTICE: TRUNCATION

$$1 \leq j \leq 100 = J$$

SOLVE NUMERICALLY THIS MATRIX EIGENVALUE PROBLEM, GET $\vec{c}^{(n)}, E_n$
 $1 \leq n \leq J$

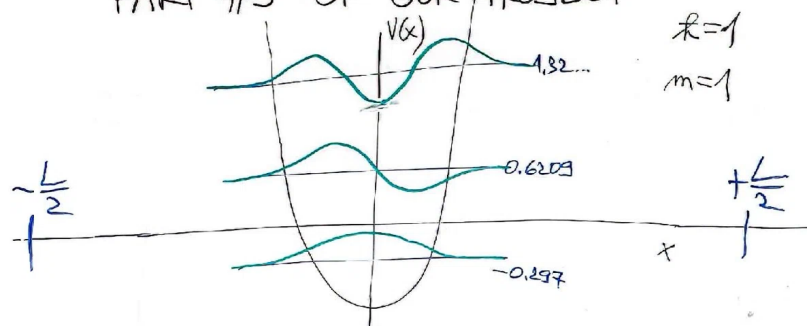
$$E_1 \leq E_2 \leq E_3 \leq \dots \leq E_J$$

NORMALIZATION OF EIGENVECTORS:

$$\vec{c}^{(n)*} \cdot \vec{c}^{(n)} = 1$$

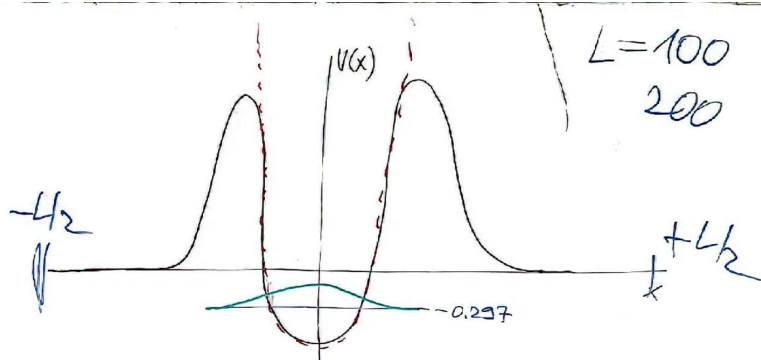
$$\psi_m(x) = \sum_{j=1}^J c_{jm} \phi_j(x)$$

PART #5 OF OUR PROJECT



$$\hbar = 1$$

$$m = 1$$



$$L = 100$$

$$200$$

$$V(x) = 0.5x^2 - 0.8$$

WE ARE GOING TO USE THE PARTICLE-IN-THE-BOX BASIS SET

$$\phi_j(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{j\pi}{L}\left(x - \frac{L}{2}\right)\right)$$

$$= 0 \quad |x| > \frac{L}{2}$$

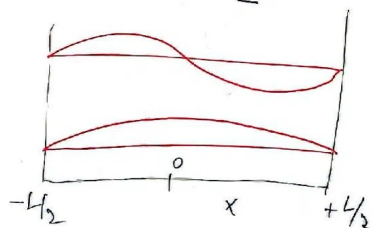
COMPLETE ON

$$L^2\left(-\frac{L}{2}, \frac{L}{2}\right)$$

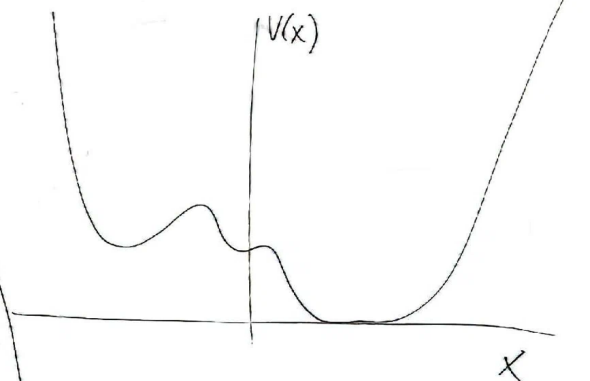
$$\hat{H} = -\frac{\hbar^2}{2m} \partial_{xx} + V(x)$$

$$\hat{H}\psi_n(x) = E_n \psi_n(x)$$

$$-\frac{L}{2} \leq x \leq \frac{L}{2}$$



$$V(x) = (0.5x^2 - 0.8)e^{-0.1x^2}$$



COMPLETE ORTHONORMAL BASIS SET ON $L^2(\mathbb{R})$

$$\{\phi_j(x)\}_{j=1}^{\infty}$$

$$\psi_n(x) = \sum_{j=1}^{\infty} c_{nj} \phi_j(x)$$

$$\langle \psi_n | \psi_n \rangle = \sum_{j=1}^{\infty} |c_{nj}|^2$$

$$\underline{H} \quad \text{MATRIX WITH ELEMENTS} \quad \langle \phi_j | \hat{H} | \phi_{j'} \rangle$$

$$\vec{c}^{(n)} = \begin{pmatrix} c_{n1} \\ c_{n2} \\ \vdots \\ c_{jn} \end{pmatrix}$$

$$\langle \phi_j | \hat{H} | \phi_{j'} \rangle = \int_{-L/2}^{+L/2} dx \phi_j(x) \left(-\frac{\hbar^2}{2m} \partial_{xx}\right) \phi_{j'}(x) + \int_{-L/2}^{+L/2} dx \phi_j(x) V(x) \phi_{j'}(x)$$

$$\underline{H} \vec{c} = E \vec{c}$$

SOLVE NUMERICALLY THIS MATRIX EIGENVALUE PROBLEM, GET $\vec{c}^{(n)}, E_n$
 $1 \leq n \leq J = 200$

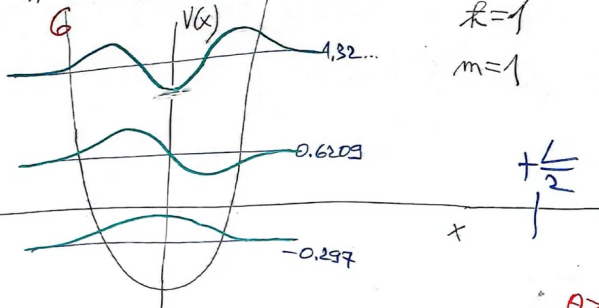
$$E_1 \leq E_2 \leq E_3 \leq \dots \leq E_J$$

NORMALIZATION OF EIGENVECTORS:

$$\vec{c}^{(n)*} \cdot \vec{c}^{(n)} = 1$$

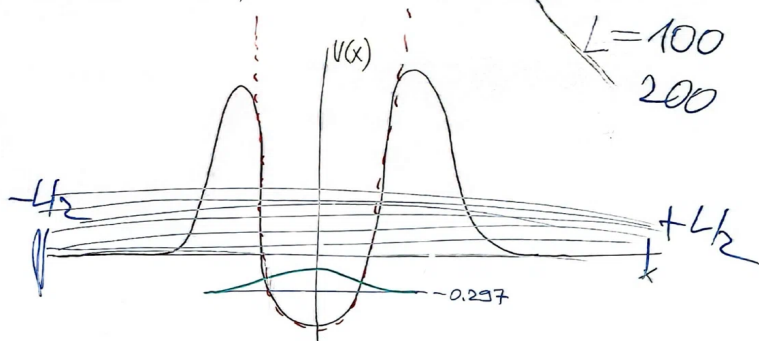
$$\psi_n(x) = \sum_{j=1}^J c_{jn} \phi_j(x)$$

PART #5 OF OUR PROJECT



$$\hbar=1$$

$$m=1$$



$$L=100$$

$$200$$

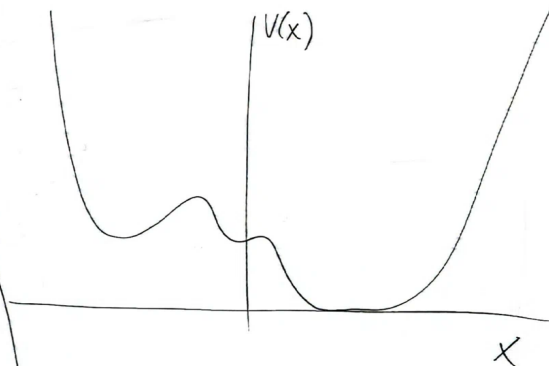
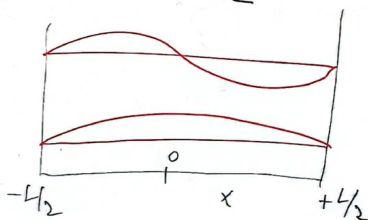
$$V(x) = 0.5x^2 - 0.8$$

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)e^{-\theta}$$

$$V(x) = (0.5x^2 - 0.8)e^{-0.1x^2}$$

$$\hat{H}\psi_n(x) = E_n \psi_n(x)$$

$$-\frac{L}{2} \leq x \leq \frac{L}{2}$$



$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}\left(x - \frac{L}{2}\right)\right)$$

$$= 0 \quad |x| > \frac{L}{2}$$

COMPLETE ON

$$L^2\left(-\frac{L}{2}, \frac{L}{2}\right)$$